2025 International Conference on Electrical, Electronics, and Computer Science with Advance Power Technologies - A Future Trends (ICE2CPT) | 979-8-3315-1499-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICE2CPT66440.2025.11340439

Power-Efficient VLSI Design: Techniques, Methodology, and HDL Implementation of a Low Power Memory Block

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***Abstract* –This paper explores techniques in Low-Power Very Large-Scale Integration (VLSI) to enhance the efficiency of digital electronics circuit design. As the demand for portable and high-performance electronic devices grows, the importance of energy-efficient solutions becomes paramount. This study examines a variety of strategies and methods designed to lower power consumption in digital circuits while preserving performance standards. It discusses essential techniques such as scaling of voltage, gating power & clock, along with innovative approaches like adiabatic computing and sub-threshold operation. Also, it covers methodology for designing low-power circuits. Additionally, the paper considers the design of a memory block and compares the power results after optimization. By providing a comprehensive analysis of current advancements and challenges in low power VLSI design, this research aims to support future initiatives in creating sustainable and efficient digital systems.**

Keywords – Low power VLSI design, CMOS, Power Dissipation, Low power strategies

I. INTRODUCTION

For Low-power Very Large-Scale Integrated (VLSI) devices in particular, power management circuit design and optimization research is essential. The semiconductor industry has made lowering power consumption a top priority as the market for portable and battery-operated devices grows. Power management circuits are essential for attaining energy efficiency because they allow electronic equipment to efficiently save energy. It takes a thorough understanding of various circuit design techniques, optimization algorithms, and energy-saving techniques to create effective power management circuits.

Over time, low power design strategies have become increasingly important. Power consumption is frequently viewed as a secondary concern in technology design, which has historically placed more emphasis on factors like performance, affordability, and dependability. However, controlling power dissipation has become crucial in today's cutthroat markets, which include wireless apps, laptops, and portable medical equipment. Depending on the application, several strategies are used to reduce power usage. The focus is on extending battery life and cutting packaging costs for battery-powered products, such as cell phones. Since electrical components in portable computers, such laptops, responsible for a significant amount of the system's total power consumption, it is imperative to minimize their power utilization. Process technology breakthroughs have made power considerations a top priority in design strategies for high-performance systems [1].

In this paper, a thorough understanding of various reasons for power loss in digital circuits is presented. Also, designing optimized power management digital VLSI circuits are main focus of this study. It looks into how different power management approach, including as gating power & clock, and scaling voltages, affect system performance, dependability, and efficiency.

II. POWER DISSIPATION SOURCE

Power dissipation, which is the rate at which energy is used up over time in electrical and electronic systems, can be divided into two types: peak power & average power. Peak power is the peak value of instantaneous power level that can occur in each period and can affect the device's reliability, possibly leading to glitches and raising the risk of system malfunctions, while average power influence the efficiency and design of cooling and packaging systems. The total power dissipated by a digital circuit can be expressed as

$$p(t) = v(t)i(t) \tag{1}$$

where v(t) is bias-voltage supplied by the power source and i(t) is instantaneous current driven by the digital circuits. Maximum instantaneous power or average power is the main focus of power minimization techniques. The primary causes of power dissipation in CMOS circuits are static, short-circuit, and dynamic power.

A. Static Power Dissipation

Static power consumption occurs when a circuit is idle, primarily due to subthreshold leakage currents and reverse-biased diode currents. Additional sources include Gate-oxide tunneling leakage and Gate-Induced Drain Leakage (GIDL). Subthreshold-leakage arises when transistors operate in a weak inversion state below the threshold voltage, which influences the magnitude of this leakage. In circuits smaller than 100 nm, subthreshold leakage is the most significant type, driven by the diffusion of minority carriers between the drain & source, even when the gate-voltage is below the threshold [2]. The subthreshold leakage current, denoted as $I_{subthreshold}$, can be mathematically defined as below

$$I_{subthreshold} = I_0 e^{\frac{V_{gs} - V_{th}}{nV_T}} \left[1 - e^{-\frac{V_{ds}}{V_T}} \right] \quad (2)$$

In Figure 1 the leakage current in a CMOS- inverter circuit is shown. The subthreshold current and junction-leakage current are also depicted below.

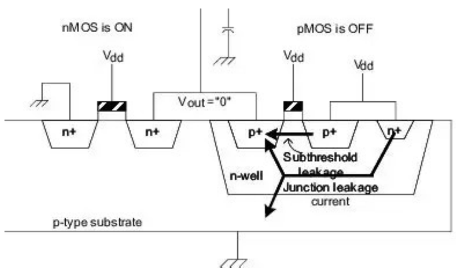

Fig 1. Static-power sources in CMOS circuit

B. Dynamic Power Dissipation

Dynamic power consumption in CMOS circuits arises from input signal changes, mainly during node charging and discharging. Although dynamic-power holds as primary source of power dissipation, leakage power is becoming more significant with smaller technologies. In CMOS-inverter, dynamic power dissipation occurs when PMOS transistor turns-on and NMOS turns-off, charging the load as the input signal falls to zero. Conversely, when the input signal rises, NMOS turns-on and PMOS turns-off, discharging the load. Figure 2 shows the dynamic power dissipation in a CMOS-inverter circuit. The mathematical expression for dynamic power dissipation in a inverter can be derived accordingly as

$$P_{dynamic} = \alpha C_L V_{DD}^2 f \quad (3)$$

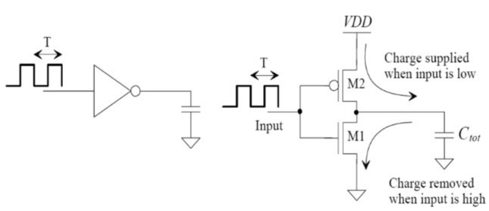

Fig 2. Dynamic power dissipation in CMOS circuit [2]

C. Short Circuit Power Dissipation

Short circuit leakage has a notable impact on dynamic power consumption. During signal transitions, the gate voltage can temporarily fall between the PMOS and NMOS thresholds, activating both transistors simultaneously. This forms a straight conductive path from voltage-source to ground, leading to power dissipation [3]. The extent of this power loss is influenced by the duration/slope of gate input and I-V characteristics of transistor's loading capacitances. Figure 3 shows short-circuit power loss as well as I/O-characteristics of a inverter.

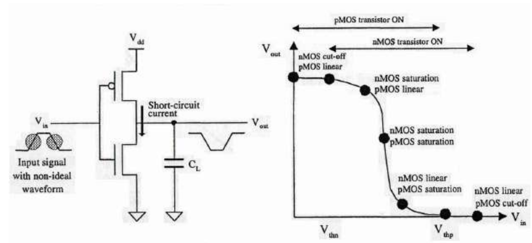

Fig 3. Short circuit power dissipation and I/O characteristics of a CMOS inverter [3]

III. LOW POWER DESIGN METHODOLOGY

The low power VLSI design methodology consist of various stages which begins at System Specification to Physical Design level power estimation.

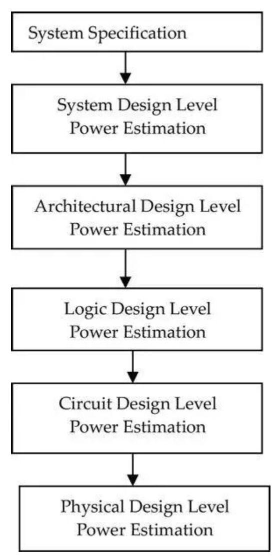

Fig 4. Flow chart for Low power VLSI design methodology

In Figure 4, flow chart consisting of various stages of low power VLSI design methodology has been shown. System Specification is the primary step in the VLSI design flow, establishing system's intended functionality, performance targets, and constraints. It creates a structured approach for the design process, specifying the system's required input/output behaviour, data processing tasks, and communication protocols. Critical performance parameters like speed, power consumption, and chip area are outlined to ensure they meet the application's demands.

System Design/Register Transfer Level (RTL) power estimation is crucial in VLSI design for optimizing power consumption early in development. At the RTL stage, designers can make architectural and micro-architectural decisions that significantly affect the chip's power efficiency. This involves assessing the design's switching activity, a key factor in dynamic power consumption. RTL power estimation tools use simulation data to predict signal toggling frequency, helping designers identify high-power components and explore power reduction approach like gating clock, operand isolation, and scaling voltages. Accurate power prediction at this stage enables informed decisions that balance power, area, and performance, leading to more efficient semiconductor products.

Architectural-level power estimation is crucial for creating energy-efficient VLSI systems. It assesses power consumption related to peripherals, interconnects, memory, and processor cores. Scenario simulation and power profile analysis help explore performance, power, and area trade-offs. Techniques like high-level modeling and analytical estimation identify bottlenecks and optimize efficiency. Focusing on this level allows designers to reduce power use through pipelining, parallelism, and data management, enhancing competitiveness and sustainability in semiconductor products.

Logic design level power estimation is vital in VLSI design, focusing on power consumption at the gate and circuit levels. Designers assess logic gates, circuits, and connections, considering static power from leakage currents and dynamic power from gate switching. Techniques such as logic simulation and power-aware synthesis enable accurate estimation, facilitating strategies like signal transition optimization and logic depth reduction to enhance power efficiency.

In VLSI design, power estimation at the circuit and physical levels focuses on transistors and layout details. Circuit-level estimation uses SPICE simulations to optimize power by considering threshold voltages, capacitance, and leakage currents. Physical-level estimation employs power grid analysis and thermal modeling to assess chip layout for effective power distribution and heat management. Techniques like layout compaction, power gating, and transistor sizing optimize power efficiency without sacrificing performance or reliability [4].

IV. POWER OPTIMIZATION TECHNIQUES

A. Transistor Level Techniques

Varying Threshold Voltage: Power consumption can be reduced by modifying MOS transistors' threshold-voltage. Static power dissipation is decreased by higher threshold voltages because they reduce subthreshold leakage. A lower supply voltage is made possible by lower threshold voltages, which lowers dynamic power dissipation.

High K dielectric material: In deep sub-micron technologies, scaling has reduced gate dielectric thickness to 2-3 atomic layers, causing significant leakage with low-K insulators like SiO_2. Industries now use high-K insulators like HfO_2 and HfSiON to reduce leakage, despite issues like mobility degradation and short-channel effects.

B. Circuit Level Techniques

Sizing transistor and pin reordering: Transistor size impacts power dissipation and gate latency in combinational circuits. Wider transistors reduce gate delay but increase switching power. Shorter channel transistors enhance performance due to higher saturation current but raise static power through increased subthreshold leakage. Additionally, input pin ordering influences power dissipation in digital circuits.

Gate Reorganization: A balanced tree structure for circuits is less prone to glitches resulting in less switching activity effectively reducing dynamic power consumption as depicted in figure 5.

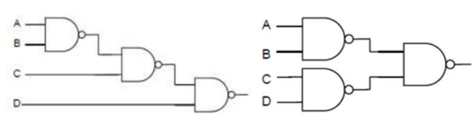

Fig 5. Balanced tree structure CMOS circuit [5]

Sleep Transistors: Sleep transistors reduce leakage currents and static power by disconnecting power from unused circuit segments. Integrating them into power domains allows designers to significantly lower overall chip power consumption by switching off inactive areas [5]. Figure 6 shows the sleep transistors working in CMOS circuits.

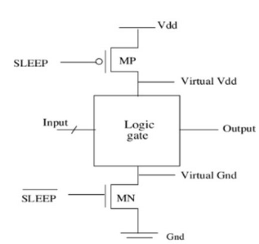

Fig 6. Sleep transistors in CMOS circuit

C. Logic Level Techniques

State Machines Encoding: In FSMs, state machine encoding optimizes power usage. Gray encoding minimizes bit transitions and dynamic power, ideal for low-power applications. One-hot encoding increases area and static power but reduces switching activity. Choosing the right encoding balances design constraints and power efficiency.

Bus Invert Encoding: To reduce switching, each clock cycle compares the next and current bits at the source. If different, the next bit is inverted, and a polarity status bit is sent to the output XOR gate to reverse the inversion [6]. This technique reduces dynamic power dissipation during high-switching data transmission on buses as shown in figure 7

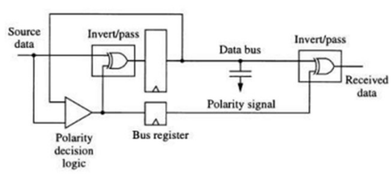

Fig 7. Bus Invert Encoding Architecture [6]

Clock Gating: Clock supply can be hold for inactive portion of circuit resulting in less switching activity leading to less power consumption as demonstrated in figure 8.

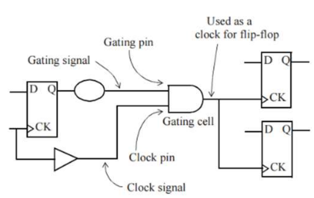

Fig 8. Clock Gating Architecture

D. Architecture Level Techniques

Parallel Architecture: In L-parallel architecture, frequency can be reduced to (f/L) without affecting throughput, allowing VDD reduction to βVDD (0 < β < 1) due to extended charging time [7]. This is shown in figure 9. Consequently, power consumption for L-stage parallel architecture is expressed as.

$$P_{parallel} = LC\,(\beta V_{DD})^2\,(f/L) = \beta^2 C V_{DD}^2 f \qquad (4)$$

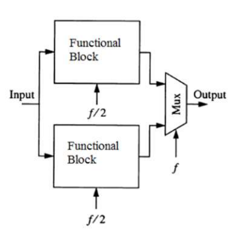

Fig 9. Two-stage Parallel Architecture [7]

Pipelining for low power: The critical path and capacitance in an N-level pipelined system are lowered to (1/N) of their initial levels. Only (1/N) of the capacitance is charged and discharged at the same clock speed, enabling the supply voltage to drop to (βVDD), where (0 < β < 1).

$$P_{pipeline} = C\beta^2 V_{DD}^2 f \qquad (5)$$

A. Advanced Techniques

Adiabatic Logic Design: Differential circuits with four-phase power clocks and variable supply voltage use cross-coupled PMOS for positive feedback and NMOS for logic evaluation. Full adiabatic logic is complex and energy-wasting, while quasi-adiabatic logic is efficient at low frequencies [8]. Adiabatic circuits are energy-efficient at low frequencies but consume energy similar to static CMOS at higher frequencies., as shown in Figure 11, highlighting a key limitation of adiabatic logic.

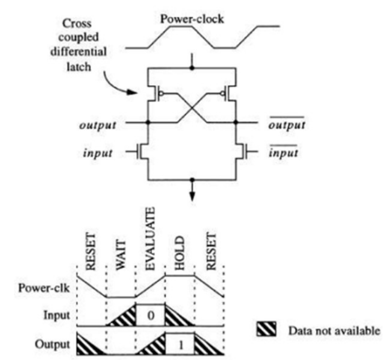

Fig 10. Four phase adiabatic inverter

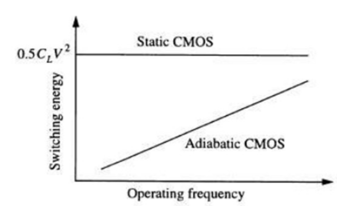

Fig 11. Switching Energy of Adiabatic Logic and Static CMOS

Asynchronous Computation: High-frequency clock signals in synchronous units cause significant dynamic power dissipation. Asynchronous units reduce power by eliminating clock signals, using acknowledge and request signals for synchronization, which minimizes signal routing and switching activity, as shown in Figure 12

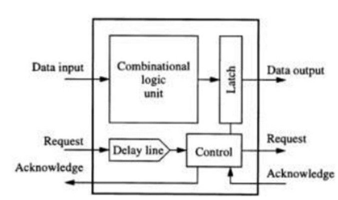

Fig 12. General asynchronous computation unit

V. DESIGN OF A LOW POWER MEMORY BLOCK

Designing a RAM block involves creating a memory structure that efficiently stores and retrieves data using address inputs. In digital design, RAM typically consists of an array of memory cells organized into words, accessible through address lines, and supports read and write operations controlled by enable signals.

Power optimization is crucial, especially for applications sensitive to energy use. Techniques to reduce power consumption include memory banking, which divides RAM into smaller sections, activating only the necessary bank during operations to save power. Gating clock reduces dynamic-power required by preventing clock signal during idle periods, preventing unnecessary switching. These methods collectively contribute to a power-efficient RAM design, balancing performance with energy conservation.

Here a RAM block has been coded in System Verilog to assess the effectiveness of power optimization techniques. First the block has been coded without any power optimization techniques. Then, the design has been optimized to implement low-power optimization techniques such as memory partitioning and gating the clock in HDL coding.

The design has been synthesized and implemented in Artix-7 Low Voltage FPGA family for which the power report is generated in Xilinx Vivado. The synthesis log in which the cell and area usage are analysed and compared is shown in figure 13.

Clock gating which allows clock to be provided only when it is performing write operation. Also, memory banking is used to partition the memory into smaller section activating only necessary section during operations. Both of the above techniques will have significant impact on switching activity within circuit thereby considerably reducing dynamic power dissipation.

As can be seen from the synthesis report of both unoptimized and optimized design that by implementing clock gating, unnecessary clock signals are removed, which can lead to a cutback in the number of flip-flops and associated logic required to manage these signals. This reduction in clock-related logic can decrease the overall cell count. Similarly, by dividing memory into banks and activating only the necessary bank can lead to a more efficient use of resources.

This selective activation can reduce the need for additional control logic, thereby decreasing the number of cells required.

```
****** Vivado v2016.2 (64-bit)
  **** SW Build 1577090 on Thu Jun  2 16:32:40 MDT 2016
  **** IP Build 1577682 on Fri Jun  3 12:00:54 MDT 2016
    ** Copyright 1986-2016 Xilinx, Inc. All Rights Reserved.

source optimized_ram.tcl -notrace
Command: synth_design -top optimized_ram -part xc7a100tlcsg324-2L
Starting synth_design
Attempting to get a license for feature 'Synthesis' and/or device 'xc7a100tl'
INFO: [Common 17-349] Got license for feature 'Synthesis' and/or device 'xc7a100tl'
INFO: Launching helper process for spawning children vivado processes
INFO: Helper process launched with PID 1708
---------------------------------------------------------------------------------
Starting Synthesize : Time (s): cpu = 00:00:06 ; elapsed = 00:00:10 . Memory (MB): peak = 305.051 ; gain = 98.285
---------------------------------------------------------------------------------
INFO: [Synth 8-638] synthesizing module 'optimized_ram' [C:/Users/HP/Desktop/paper_new_optimized_ram/paper_new_optimized_ram.srcs/sources_1/new/optimized_ram.sv:23]
```

```
---------------------------------------
Start Writing Synthesis Report
---------------------------------------

Report BlackBoxes:
+-+--------------+----------+
| |BlackBox name |Instances |
+-+--------------+----------+
+-+--------------+----------+

Report Cell Usage:
+------+------+------+
|      |Cell  |Count |
+------+------+------+
|1     |BUFG  |     1|
|2     |LUT5  |    16|
|3     |LUT6  |    32|
|4     |MUXF7 |    16|
|5     |MUXF8 |     8|
|6     |FDCE  |   128|
|7     |IBUF  |    15|
|8     |OBUF  |     8|
+------+------+------+

Report Instance Areas:
+------+---------+-------+------+
|      |Instance |Module |Cells |
+------+---------+-------+------+
|1     |top      |       |   224|
+------+---------+-------+------+
```

```
---------------------------------------
Start Writing Synthesis Report
---------------------------------------

Report BlackBoxes:
+-+--------------+----------+
| |BlackBox name |Instances |
+-+--------------+----------+
+-+--------------+----------+

Report Cell Usage:
+------+------+------+
|      |Cell  |Count |
+------+------+------+
|1     |BUFG  |     1|
|2     |LUT2  |     1|
|3     |LUT5  |     8|
|4     |LUT6  |    16|
|5     |MUXF7 |     8|
|6     |FDCE  |    64|
|7     |IBUF  |    15|
|8     |OBUF  |     8|
+------+------+------+

Report Instance Areas:
+------+---------+-------+------+
|      |Instance |Module |Cells |
+------+---------+-------+------+
|1     |top      |       |   121|
+------+---------+-------+------+
```

Fig 13. Synthesis Report of unoptimized and optimized power aware design of memory block

Figures 14 and 15 display the synthesized netlist and power report summary, detailing dynamic and static power consumption for an unoptimized design

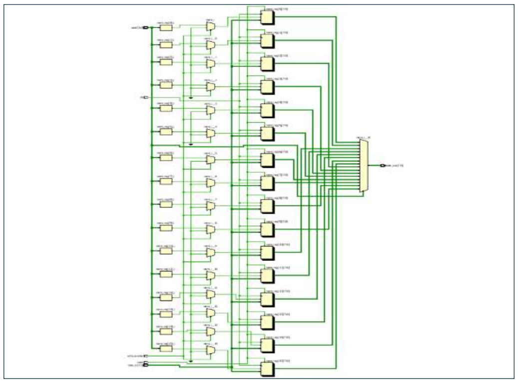

Fig 14. Synthesized netlist of simple memory block

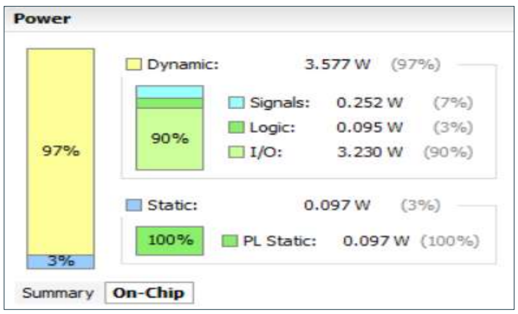

Fig 15. Power summary report of simple memory block

Figures 16 and 17 display the synthesized netlist and power report summary for an optimized design.

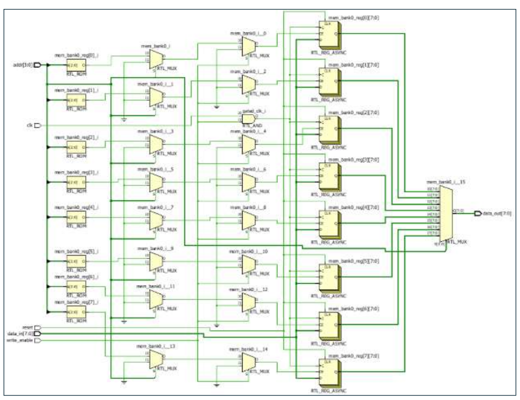

Fig 14. Synthesized netlist of power optimized memory block

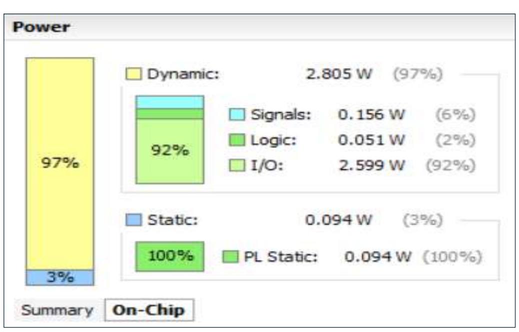

Fig 14. Power report of optimized memory block

VI. CONCLUSION & FUTURE SCOPE

In this paper, various low power design methodology has been discussed along with the optimized low power design implementation of a memory block. The various sources of power dissipation have been presented along with low power design methodology for efficient digital circuit design which covers from System specification to Physical design level power estimation. This paper also covers various power optimization techniques ranging from transistors level, circuit level, logic level, architecture level and some advanced techniques. At the end, a comparison has been shown in the power numbers and synthesis results for a memory block design between simple and low power optimized design implementations. As future work, the above optimization techniques can be implemented on more complex design blocks such as ALU, protocols etc. to access the efficiency of the above techniques.

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